

KOUCHIBOUGUAC, NB, Canada

WMO: 716700

Lat: 46.789N

Lon: 65.013W

Elev: 34

StdP: 100.92

Time Zone: -4.00 (NAA)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-19.8	-26.6	0.3	-21.1	-24.1	0.4	-18.3	6.5	-5.3	6.3	-6.2	1.5	260	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.2	21.0	28.5	20.1	26.8	19.2	22.5	27.7	21.6	26.2	20.7	24.9	2.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.6	24.7	20.0	14.8	24.1	19.1	14.0	23.1	66.4	27.9	63.1	25.8	59.8	25.1	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.2	4.3	-26.5	33.6	2.2	1.5	-28.1	34.6	-29.4	35.5	-30.6	36.3	-32.2	37.3		
(5)			-26.7	24.2	2.2	0.7	-28.2	24.7	-29.5	25.1	-30.8	25.5	-32.4	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-8.0	-7.4	-2.6	3.9	10.2	15.6	20.2	19.2	14.7	8.2	1.8	-4.6
	DBStd	10.83	6.40	5.73	5.20	4.00	4.29	3.97	3.22	2.94	4.09	4.10	5.07	5.58
	HDD10.0	2469	557	486	393	189	49	3	0	0	6	85	249	452
	HDD18.3	4659	815	719	651	434	255	102	18	25	121	314	495	710
	CDD10.0	1016	0	0	1	6	56	170	316	286	148	29	4	0
	CDD18.3	164	0	0	0	0	3	18	76	53	13	1	0	0
	CDH23.3	1465	0	0	2	4	68	216	654	411	106	4	0	0
	CDH26.7	391	0	0	0	0	17	59	197	94	24	0	0	0
(14) Wind	WSAvg	1.7	1.8	2.0	2.1	1.8	1.7	1.5	1.3	1.3	1.4	1.5	1.8	1.9
(15) Precipitation	PrecAvg	1158	98	80	95	84	98	92	103	83	95	111	104	106
	PrecMax	1459	153	140	170	130	168	170	187	164	198	194	159	231
	PrecMin	893	42	31	36	34	29	32	36	32	32	31	60	32
	PrecStd	129	29	27	33	26	37	29	35	32	41	44	31	40
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	9.1	15.9	22.5	29.0	31.7	33.1	31.2	29.6	23.6	18.0	11.0
		MCWB	9.5	7.6	8.8	12.6	18.3	19.7	22.0	21.6	21.3	17.9	16.9	10.3
	2%	DB	6.1	5.4	9.7	17.0	25.0	28.3	30.6	29.1	26.1	19.5	14.9	7.4
		MCWB	5.0	3.6	4.6	10.9	15.9	18.9	21.5	20.8	19.3	15.0	13.4	6.6
	5%	DB	3.0	2.8	6.8	14.1	21.6	25.8	28.9	27.4	23.9	17.1	12.3	4.9
		MCWB	2.1	0.9	3.5	9.1	14.3	17.7	20.6	19.8	18.3	14.4	10.4	4.1
	10%	DB	0.8	0.6	4.5	11.3	18.5	23.5	27.1	25.7	21.8	15.1	9.7	2.5
		MCWB	0.0	-0.3	1.6	7.0	12.6	16.8	19.8	19.1	17.4	12.5	7.9	1.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	7.7	11.3	15.1	19.3	22.0	23.9	23.3	22.3	18.3	17.1	10.4
		MCDB	10.1	8.7	15.6	19.7	25.4	27.4	29.8	28.3	27.6	21.0	18.0	11.0
	2%	WB	5.4	3.9	6.1	12.4	17.1	20.5	22.8	22.1	20.9	16.5	13.8	6.8
		MCDB	5.9	5.3	8.3	15.2	23.0	25.6	28.4	26.8	24.0	18.4	14.6	7.4
	5%	WB	2.1	1.2	3.8	9.8	15.3	19.1	21.9	21.2	19.6	14.7	10.8	4.3
		MCDB	2.9	2.1	6.5	13.2	19.3	23.7	26.8	25.5	22.7	16.7	11.9	4.9
	10%	WB	0.2	-0.1	2.0	7.4	13.4	17.6	21.0	20.2	18.2	12.7	7.9	1.9
		MCDB	0.7	0.6	4.1	10.7	17.8	21.9	25.4	24.0	21.4	14.7	9.4	2.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.5	10.6	10.7	12.3	11.6	11.7	11.4	11.6	9.7	8.4	7.8
		MCDBR	11.4	11.4	13.1	16.4	18.0	16.1	14.0	13.8	13.6	12.5	11.3	9.0
		MCWBR	11.2	9.8	9.3	10.7	9.9	8.3	6.7	7.0	8.0	8.8	9.9	8.9
	5% WB	MCDBR	11.6	10.8	11.9	14.8	15.0	13.3	11.7	11.6	11.7	10.4	10.8	9.3
		MCWBR	11.6	9.8	9.3	10.7	9.4	7.8	6.3	6.6	7.8	8.4	10.4	9.3
(40) Clear-Sky Solar Irradiance	taub		0.286	0.282	0.319	0.350	0.368	0.388	0.413	0.391	0.357	0.327	0.302	0.282
	taud		2.408	2.378	2.383	2.406	2.411	2.348	2.319	2.371	2.453	2.534	2.549	2.500
	Ebn at Noon		812	897	908	906	898	877	848	854	856	836	796	782
	Edh at Noon		69	87	101	109	113	121	123	112	94	75	61	57
(44) All-Sky Solar Radiation	RadAvg		1.37	2.11	3.33	4.22	5.01	5.41	5.51	4.82	3.81	2.39	1.32	1.00
	RadStd		0.13	0.22	0.22	0.40	0.51	0.41	0.39	0.32	0.34	0.22	0.16	0.10

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page